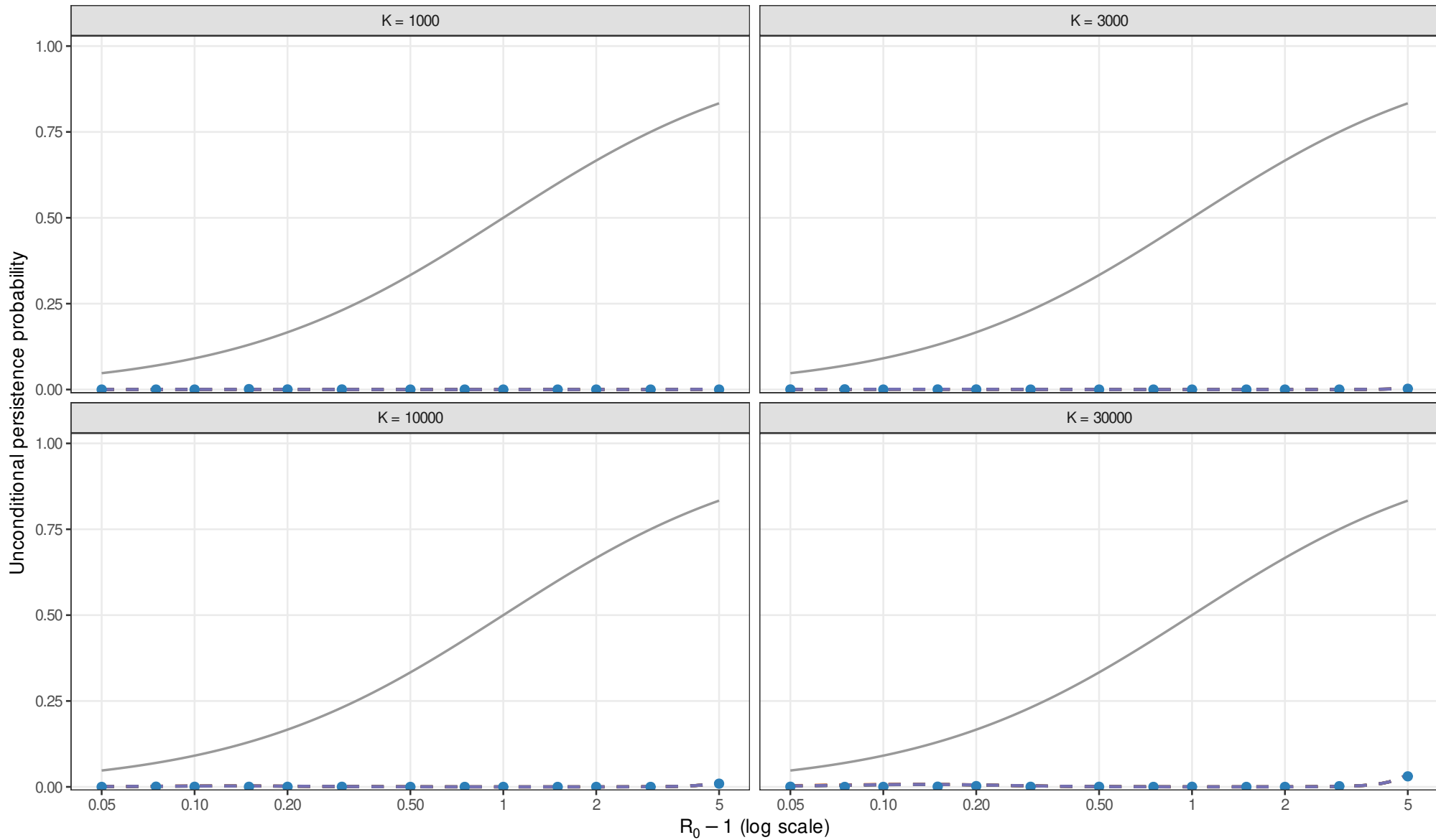


Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.01, theta = 0; l0 = 1; matching height = 3/4 compromise



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

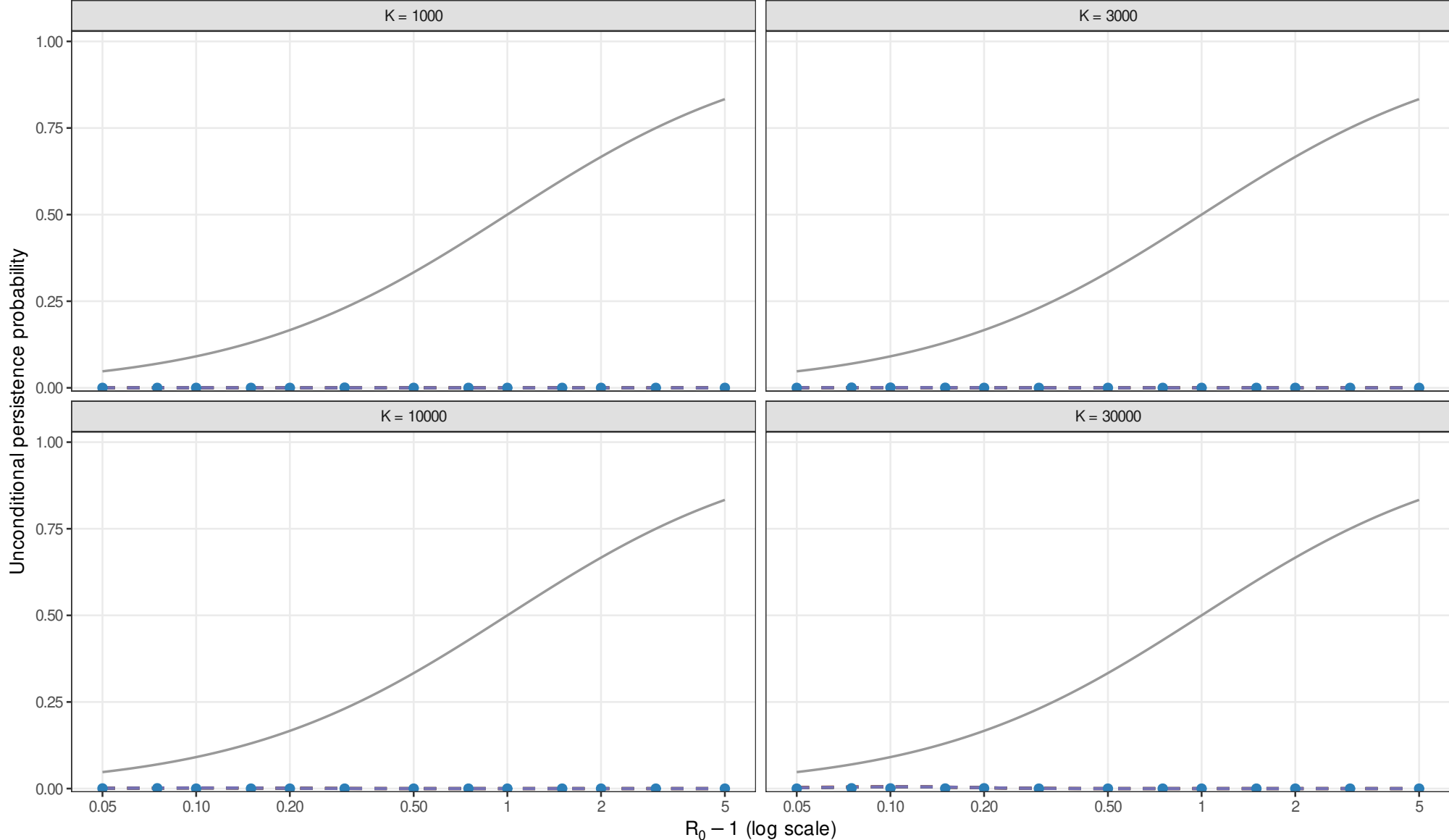
Matching height

- $y_{BL} = K^{-1/4} y_{star}^{3/4}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.01, theta = 0.5; I0 = 1; matching height = 3/4 compromise



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

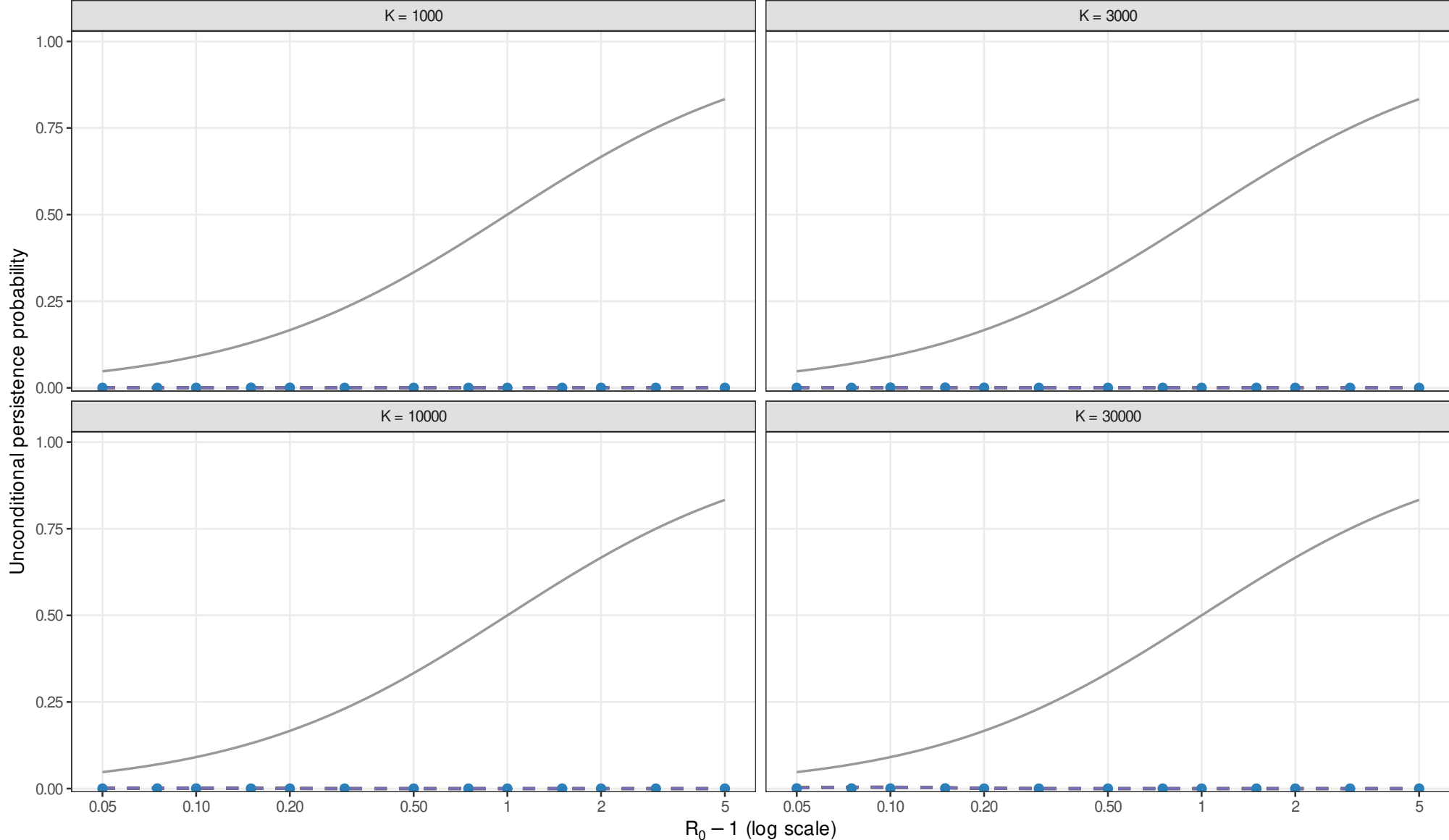
Matching height

- $y_{BL} = K^{-1/4} y_{star}^{3/4}$
- Early-establishment reference

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Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.01, theta = 1; l0 = 1; matching height = 3/4 compromise



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

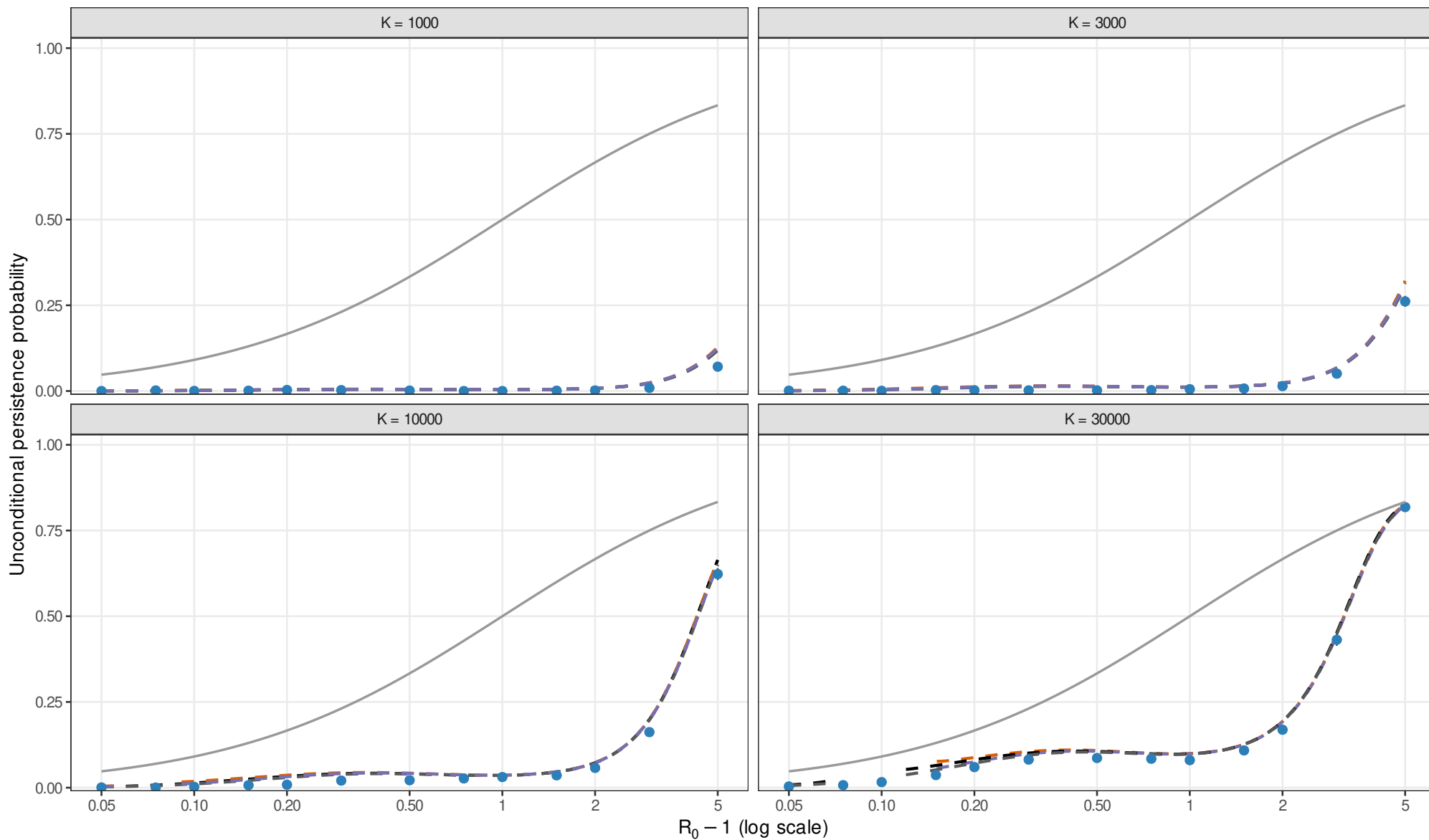
Matching height

- $y_{BL} = K^{-1/4} y_{star}^{3/4}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.02, theta = 0; l0 = 1; matching height = 3/4 compromise



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

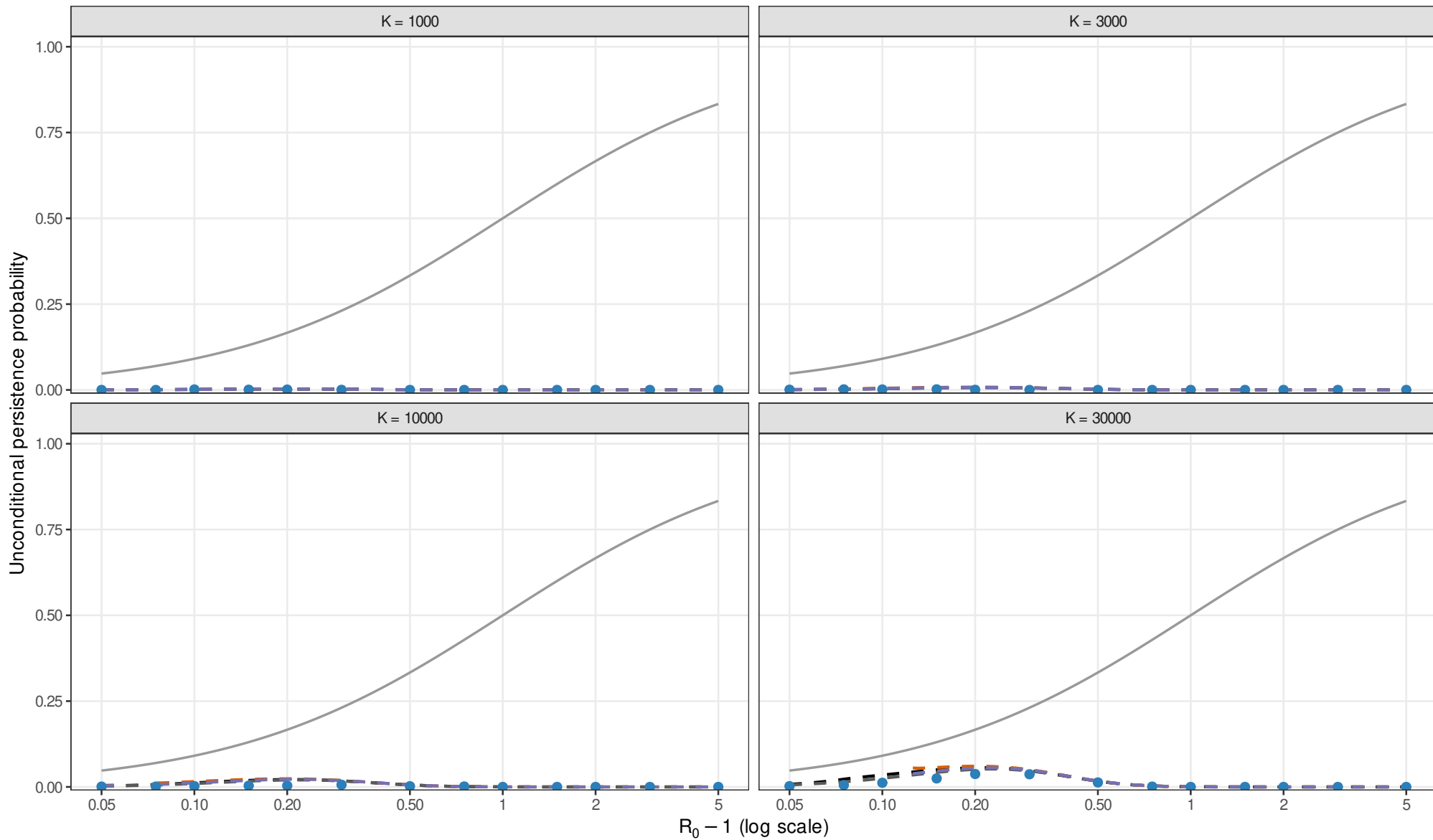
Matching height

- $y_{BL} = K^{-1/4} y_{star}^{3/4}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.02$, $\theta = 0.5$; $I_0 = 1$; matching height = $3/4$ compromise



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

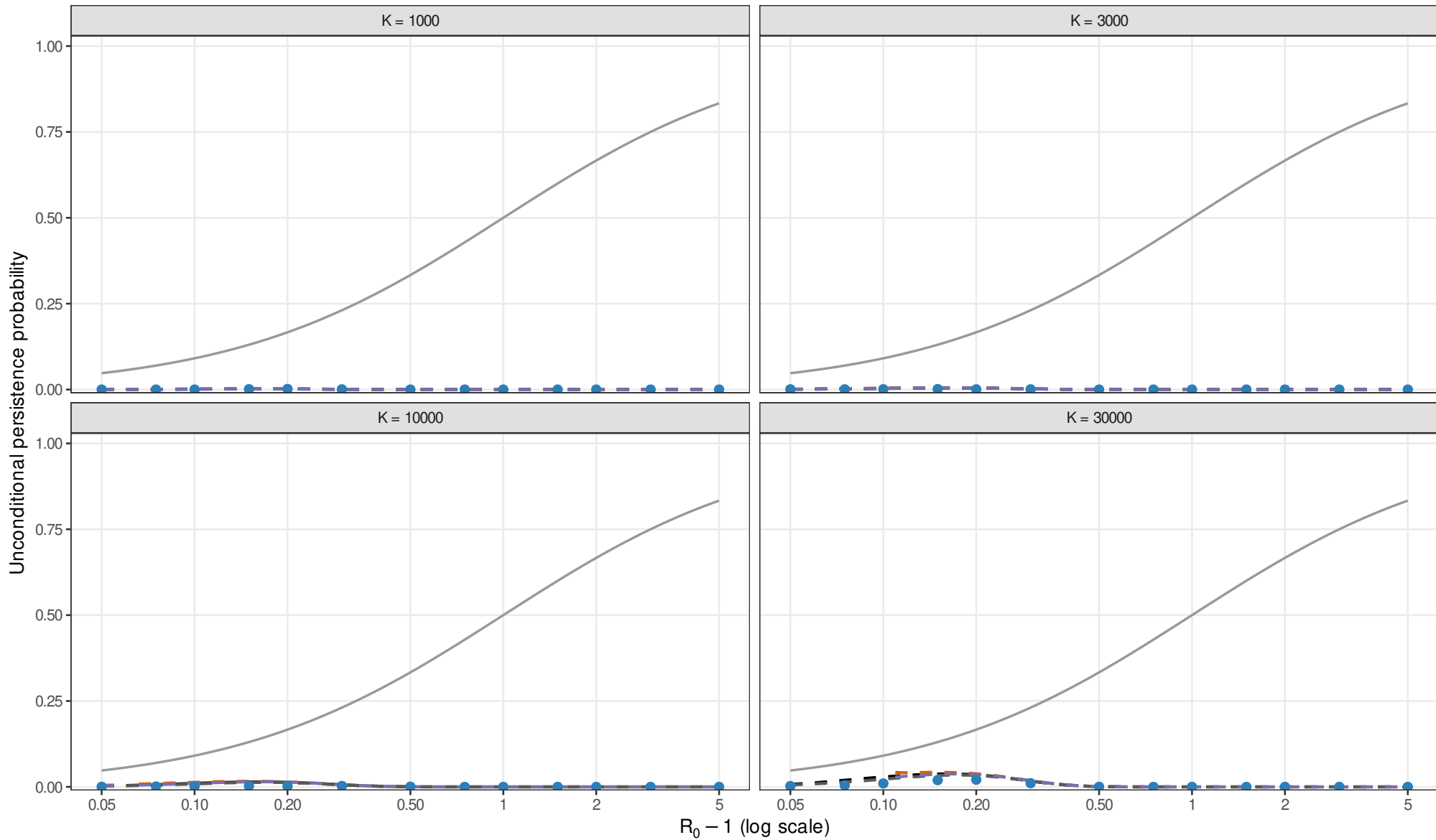
Matching height $y_{BL} = K^{-1/4} y_{star}^{3/4}$

- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height $3/4$ compromise. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.02, theta = 1; l0 = 1; matching height = 3/4 compromise



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

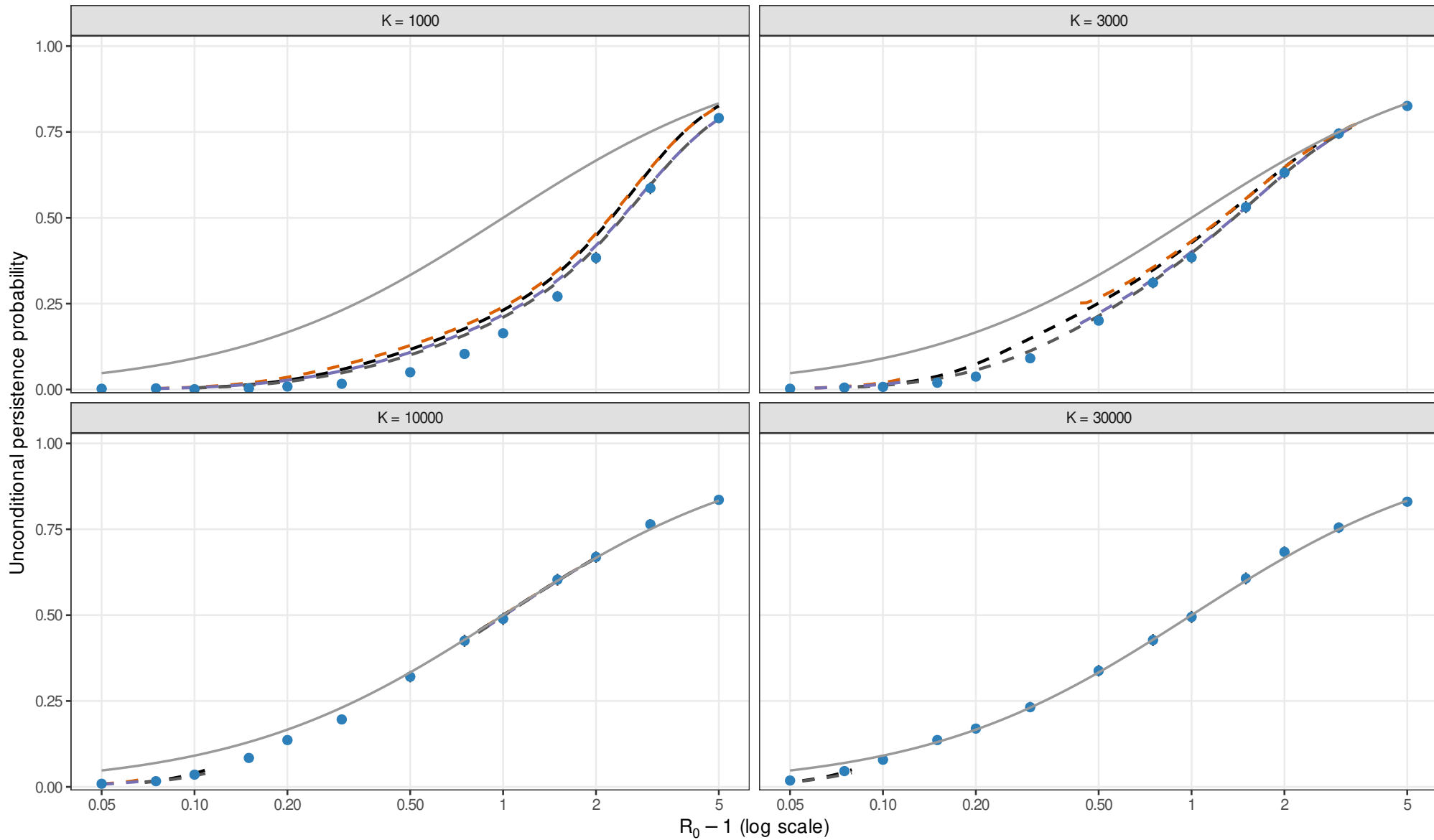
Matching height

- $y_{BL} = K^{-1/4} y_{star}^{3/4}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.05, theta = 0; l0 = 1; matching height = 3/4 compromise



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

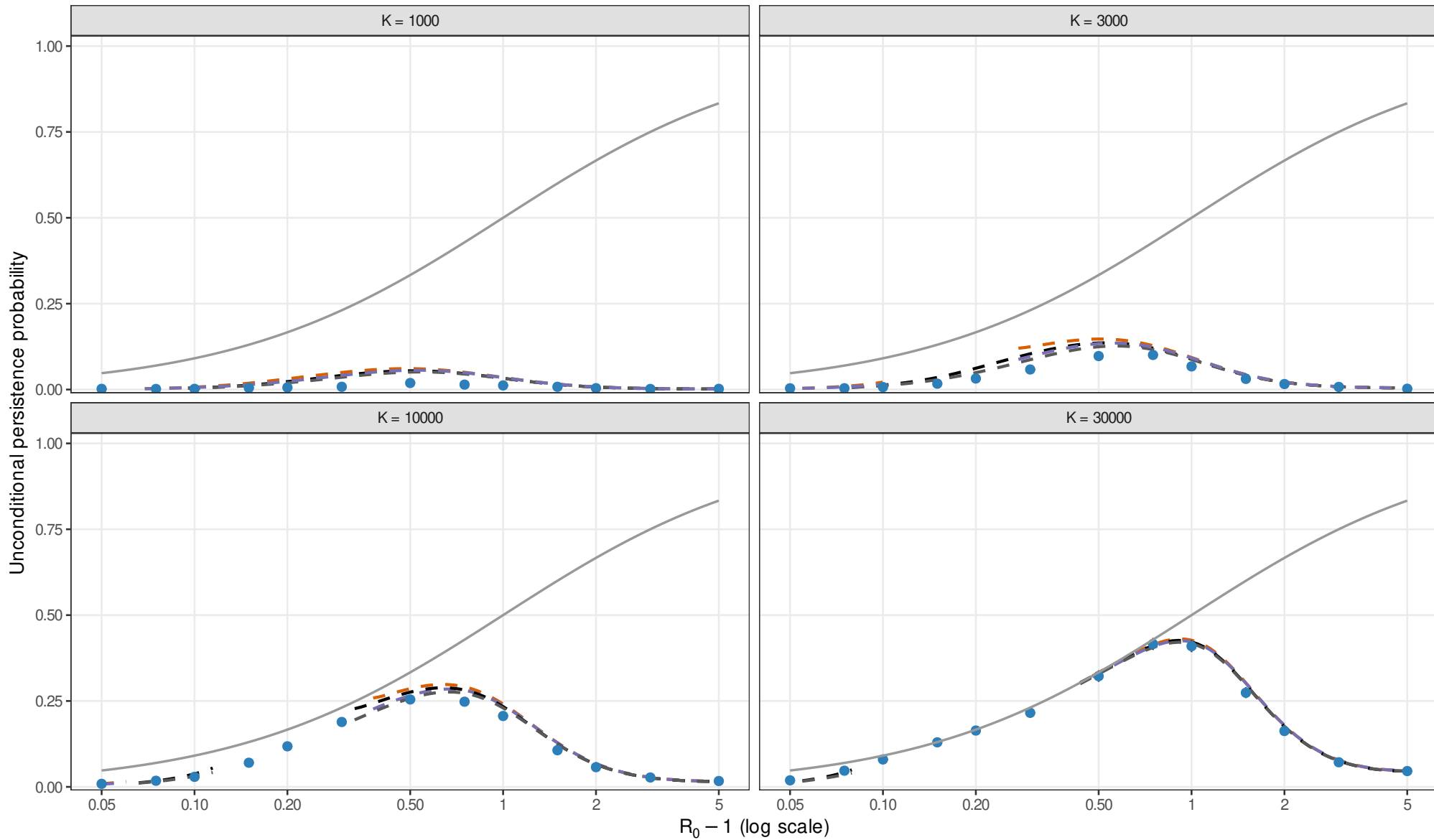
Matching height

- $y_{BL} = K^{-1/4} y_{star}^{3/4}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

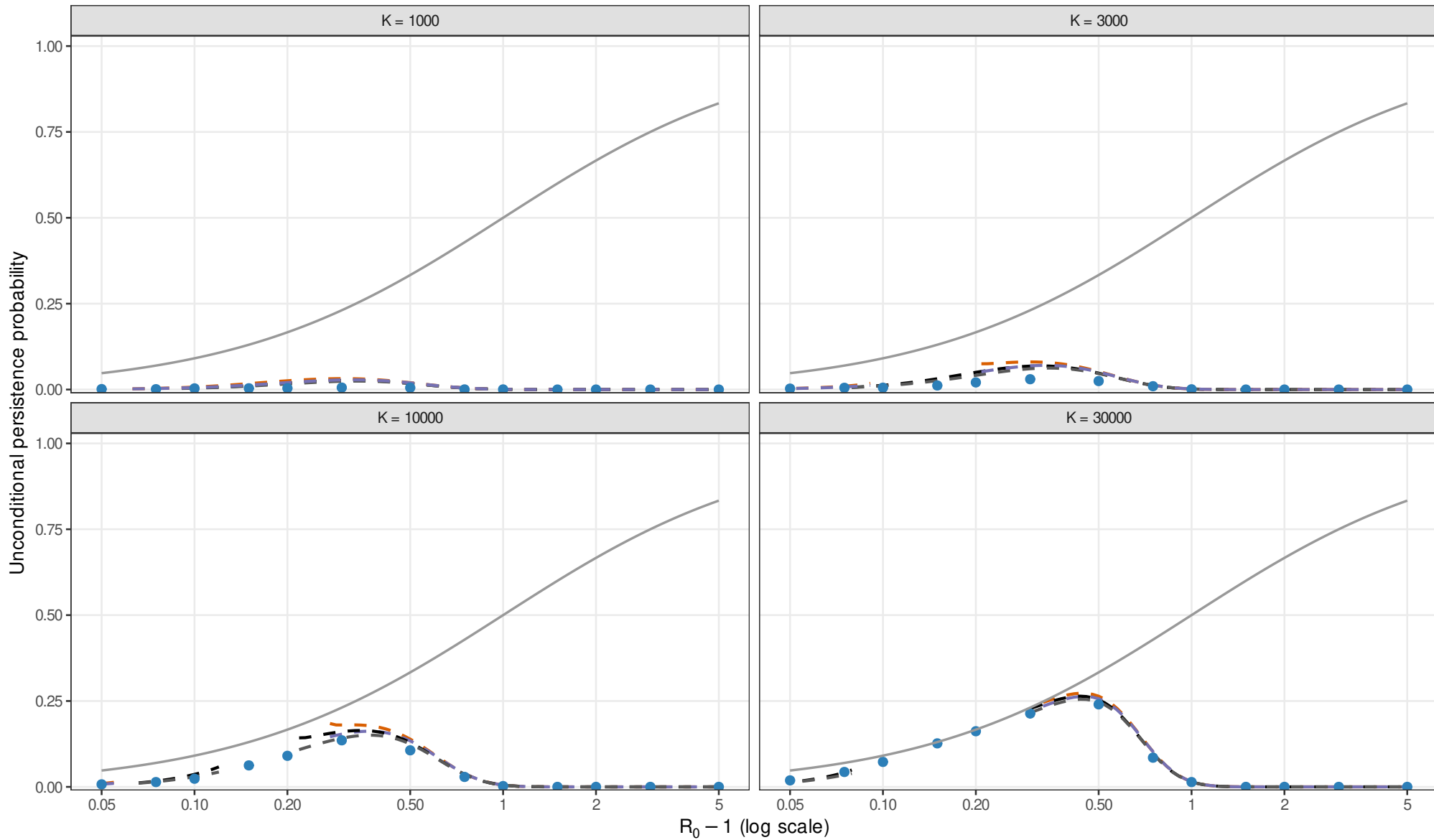
Exact Gillespie CTMC; rho = 0.05, theta = 0.5; I0 = 1; matching height = 3/4 compromise



Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; rho = 0.05, theta = 1; l0 = 1; matching height = 3/4 compromise



Analytical method

- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

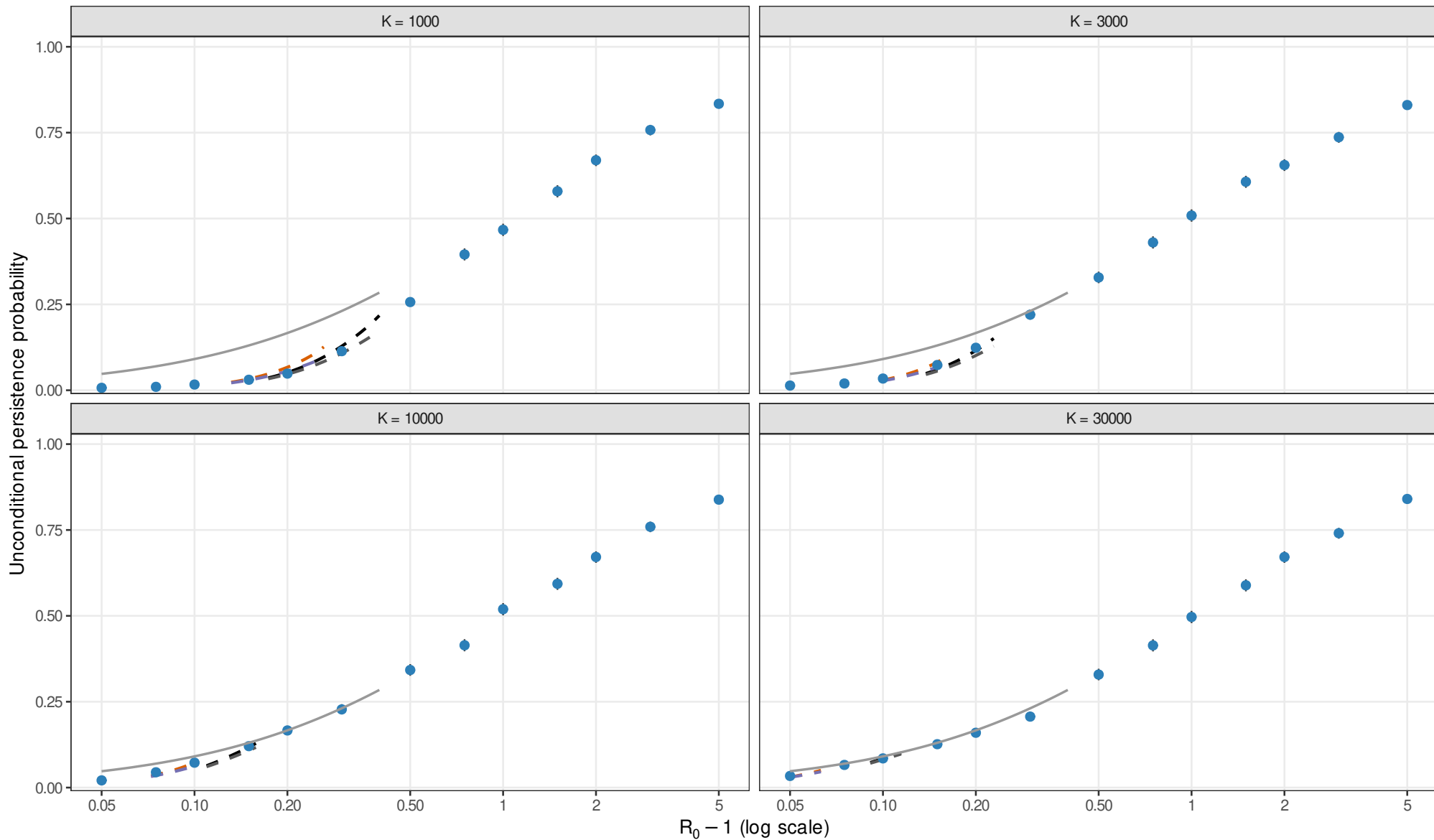
Matching height

- $y_{BL} = K^{-1/4} y_{star}^{3/4}$
- Early-establishment reference

Points and 95% Wilson intervals are stochastic estimates. All analytical curves use matching height 3/4 compromise. Each curve stops where its matching equation has no admissible root.

Full stochastic validation of unconditional persistence

Exact Gillespie CTMC; $\rho = 0.10$, $\theta = 0$; $l_0 = 1$; matching height = $3/4$ compromise



Analytical method

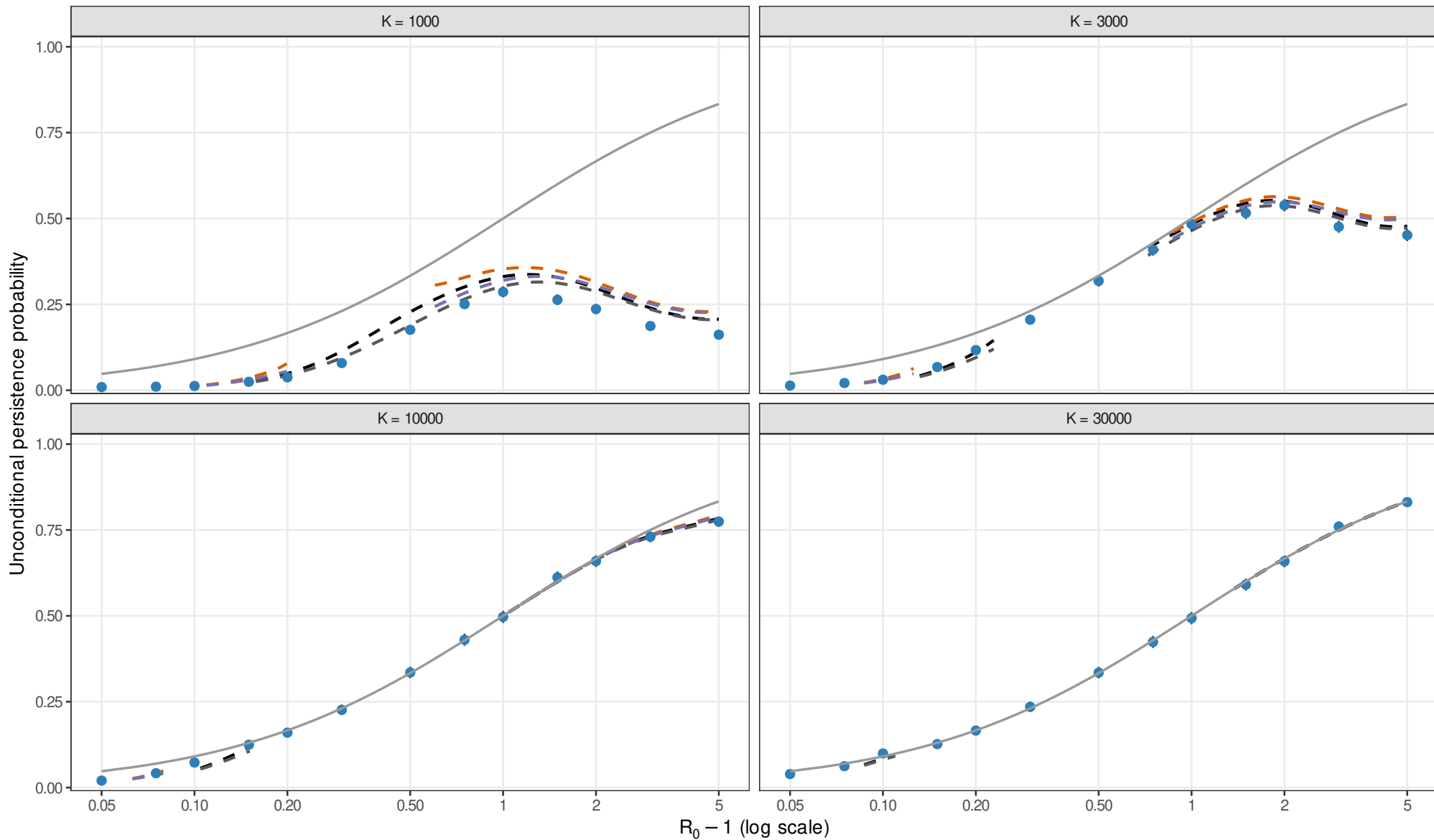
- Exact Kendall + first-order entry
- Exact Kendall + second-order entry
- Laplace + first-order entry
- Laplace + second-order entry
- Early-establishment reference

Matching height $y_{BL} = K^{-1/4} y_{star}^{3/4}$ Early-establishment reference

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Full stochastic validation of unconditional persistence

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Analytical method

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- Laplace + second-order entry
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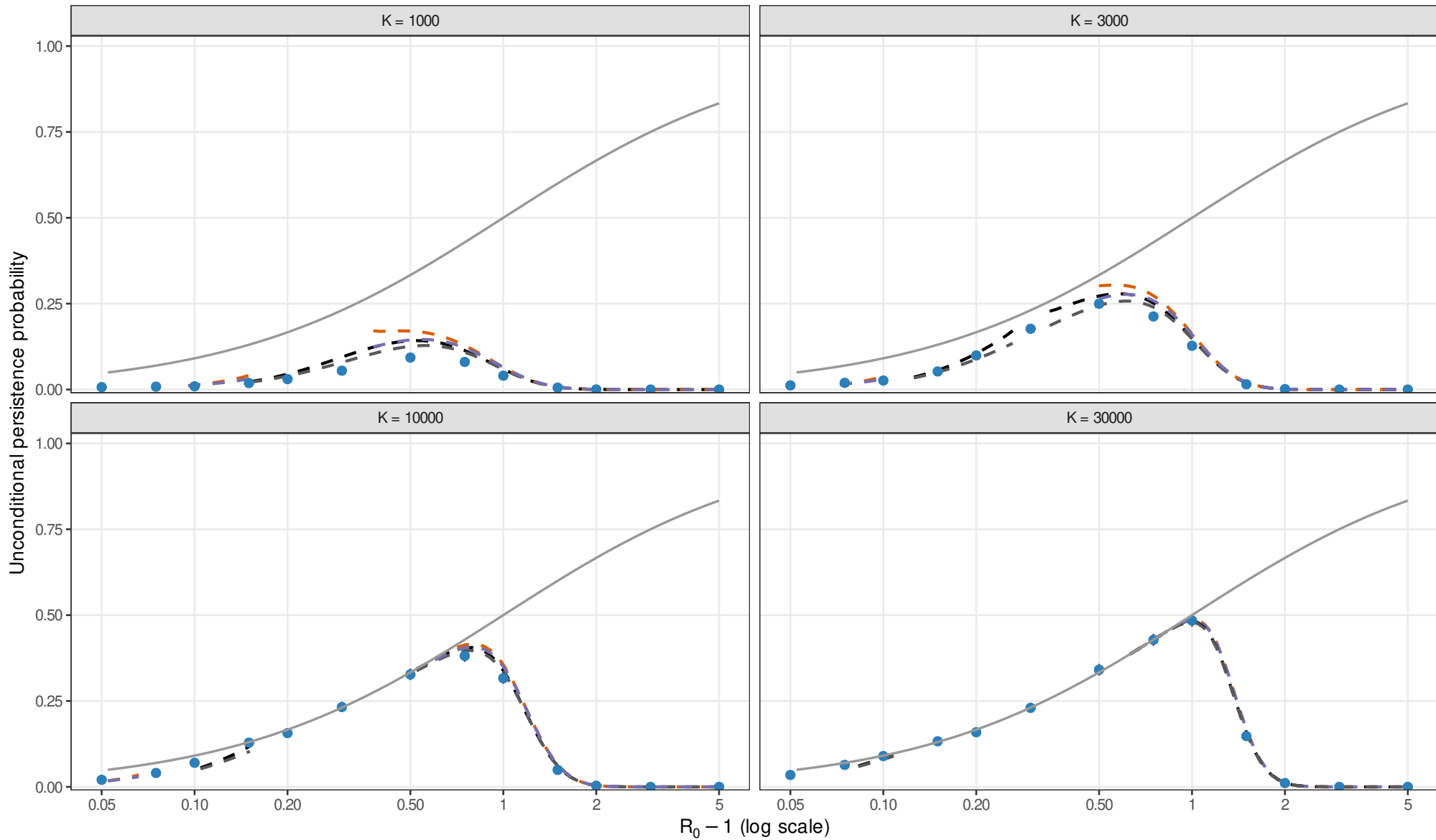
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